NightReID: A Large-Scale Nighttime Person Re-Identification Benchmark Technical Appendix

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1 The NightReID Dataset

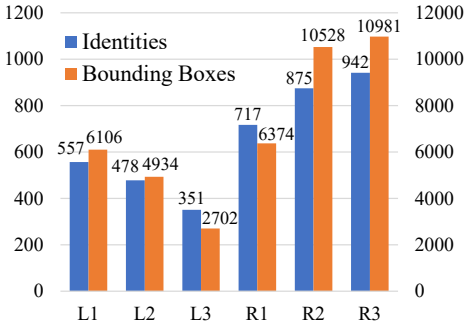
Dataset Statistics

Figure 1 presents key statistics of the NightReID dataset. Each annotated identity is captured by at least two cameras and up to six cameras, with an average of 27.6 bounding boxes per identity. In contrast, each identity in the distractors contains fewer images, leading to more varied interference. Notably, over 98% of annotated identities include at least 10 bounding boxes, and the distribution of identities and bounding boxes across different cameras is relatively balanced, facilitating robust training and evaluation. Table 1 details the statistics of the four evaluation protocol variants of the NightReID dataset.

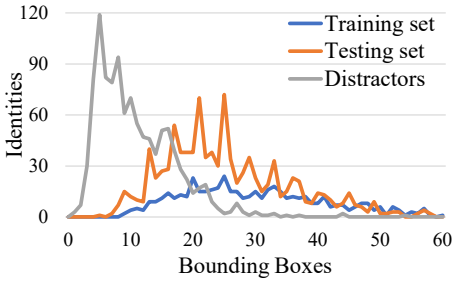
Comparison with the Night600 Dataset

As discussed in Section 3, both Night600 (Lu et al. 2023b) and NightReID are nighttime RGB Re-ID datasets, with the primary distinction being image quality. Figure 2 underscores the marked differences in pixel value distributions between the two datasets, highlighting this critical disparity.

In extremely low-light nighttime conditions, RGB (visible) cameras tend to produce dark, poor-quality images that are difficult for both human observers and computer vision models to interpret. Night600 suffers from such quality issues, leading to a poor representation of individuals in nighttime settings and significantly inferior Re-ID performance. For example, methods like TransReID (He et al. 2021) that employ standard Vision Transformer (Dosovitskiy et al. 2021) architectures require considerably more epochs to converge on the Night600 dataset, highlighting its inadequacy for effectively evaluating the performance of existing methods in nighttime Re-ID tasks. However, this challenge is not insurmountable. As illustrated in Figure 3, NightReID achieves superior image quality and richer color details under similar nighttime conditions through exposure gain technology, offering new insights into the effective use of RGB cameras for Re-ID at night and providing a valuable opportunity for comprehensive benchmarking of various methods in nighttime Re-ID tasks.



(a) The number of identities and bounding boxes captured by each camera.



(b) The distribution of identities across different bounding boxes.

Figure 1: Statistics of the NightReID dataset.

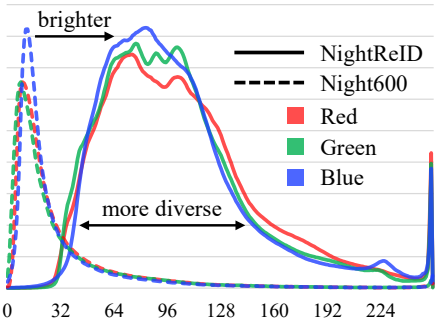


Figure 2: The pixel value distribution of each dataset.

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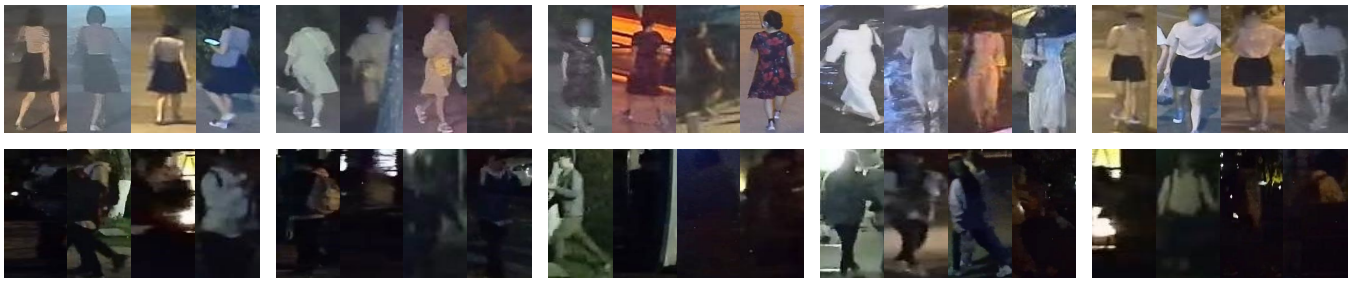
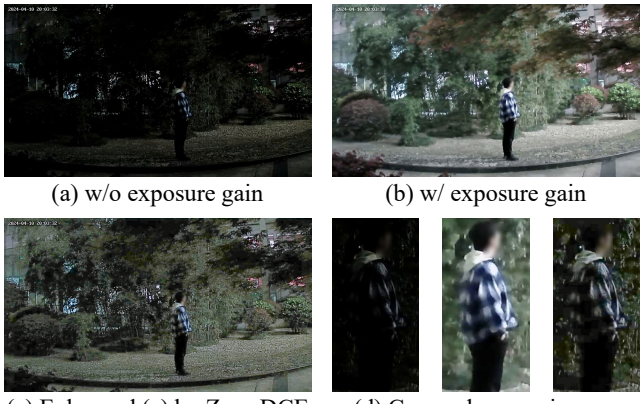


Figure 3: Comparison of image quality between NightReID and Night600. The first row represents sample images from the NightReID dataset, and the second row represents those from the Night600 dataset. Each cluster represents the same identity.

Evaluation Protocol	Query Set		Gallery Set		Distractor Set	
	#Identities	#Images	#Identities	#Images	#Identities	#Images
528 IDs w/ D	528	1394	528	14300	1096	11811
528 IDs w/o D					-	-
1000 IDs w/ D	1,000	2637	1000	25914	1096	11811
1000 IDs w/o D					-	-

Table 1: Statistics of the four evaluation protocol variants of the NightReID dataset.



(a) w/o exposure gain (b) w/ exposure gain

(c) Enhanced (a) by Zero-DCE (d) Cropped person images

Figure 4: Comparison between exposure gain and LLIE.

Cameras capture images by processing raw signals through an Image Signal Processor (ISP), which generates the final output. In low-light environments such as nighttime, brightness and contrast can be significantly enhanced by utilizing ISP signal processing and adjusting exposure settings, ensuring consistent image quality across varying lighting conditions. This technique, known as exposure gain, operates on the more fundamental continuous data at the camera's signal level, offering superior enhancement compared to Low Light Image Enhancement (LLIE) methods that process quantized RGB images. Figure 4 demonstrates this difference, where (a) represents the image without exposure gain, (b) represents the image after enabling exposure gain, and (c) represents the image enhanced by Zero-DCE (Guo et al. 2020) from (a). The three images in (d) show cropped person images from (a), (b), and (c), respectively. It is evident that (b) exhibits superior image quality and more accurate color representation compared to (c), thereby enabling downstream computer vision tasks such as Re-ID to interpret the images more effectively.

Exposure gain is a widely implemented feature among camera manufacturers, offering a cost-effective solution for capturing higher-quality data by RGB cameras during nighttime. Given NightReID's remarkable performance improvements over Night600, we strongly recommend activating exposure gain for surveillance systems during nighttime, as it maximizes the potential of RGB cameras and significantly boosts Re-ID performance.

2 Experiment

Multi-Dataset Evaluation

To demonstrate the significance and effectiveness of NightReID as a real-world nighttime Re-ID dataset, we conducted multi-dataset experiments following the approach of CENet (Lu et al. 2023a). CENet incorporates an additional synthetic dataset Syn_Dark for multi-domain learning, which simulates nighttime environments by processing the Market-1501 (Zheng et al. 2015) and MSMT (Wei et al. 2018) datasets into paired training data containing 4,852 identities and 138,984 images. In contrast, NightReID is a real-world dataset captured at night, featuring 500 identities and 15,514 images in the training set. For our experiments, we integrated only the DDA module into a single-branch architecture, normalizing each dataset with its respective statistical priors. As shown in Table 2, incorporating NightReID into the training data significantly improves performance on the Night600 dataset. Notably, NightReID achieves competitive results compared to Syn_Dark, despite having only about one-tenth of the data scale and lacking paired ground truth. This is attributed to NightReID's inclusion of various real-world nighttime degradation factors and

Method	Training data		Rank-1	mAP
CENet	Night600 300/14,462	Syn_Dark 4852/138,984	25.0	**13.3**
Ours	Night600 300/14,462	**NightReID** **500/15,514**	**26.2**	12.4

Table 2: Multi-dataset training. The Training data column lists the number of identities and images in each dataset. Both methods are evaluated on the Night600 dataset.

Architecture	Rank-1	mAP
IED only	67.0	51.7
IED to DDA	66.5	51.0
IED to BN	65.3	50.9
IED to APN	65.1	50.6
Parallel (EDA)	**67.6**	**52.3**

Table 3: The performance of different architectural designs on the NightReID dataset. The middle section presents sequential designs where the IED module is followed by the DDA module, Batch Normalization (BN), and Adaptive Parameter Normalization (APN), respectively.

its superior image quality, which synthetic datasets struggle to simulate effectively. Additionally, incorporating Night600 into training posed challenges due to its extremely low visibility, leading to suboptimal results when testing on the NightReID dataset. This suggests that excessively dark images may hinder rather than assist nighttime Re-ID learning. Consequently, we believe NightReID has the potential to drive substantial progress in nighttime Re-ID research.

Architecture Discussion

To fully leverage the complementary strengths of the proposed modules, we explored various integration strategies within the framework. Specifically, we investigated sequential architectures, where the two modules are applied consecutively, and a parallel architecture, where the two modules operate simultaneously in separate branches. In the sequential architectures, the IED module is positioned earlier in the pipeline due to its dependence on raw images for enhancement. However, as shown in Table 3, all sequential designs exhibit performance degradation compared to using the IED module alone. This degradation primarily arises because the IED module modifies the data distribution during training by enhancing the images, which creates a mismatch with the DDA module that relies on statistical priors derived from the original dataset.

To address this issue, we adopt a multi-branch parallel architecture in which the IED and DDA modules operate independently, forming the Enhancement, Denoising, and Alignment (EDA) framework. This design enhances the data at both the image and distribution levels simultaneously, enabling the model to harness the strengths of both modules effectively and achieve superior performance.

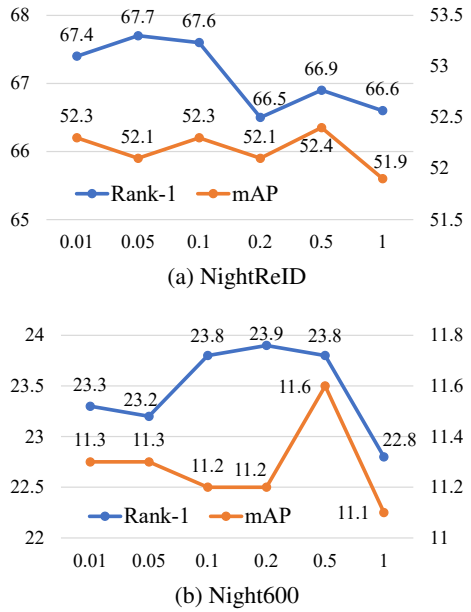


Figure 5: Hyper-parameter analysis of W_{IED}.

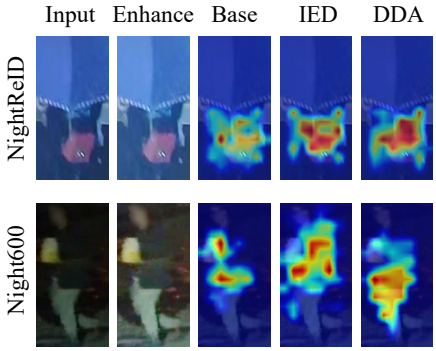


Figure 6: Visualization of activation maps.

Hyper-Parameter Analysis

To achieve the goal of extracting more discriminative features and considering that the unsupervised loss $\mathcal{L}_{IED}$ cannot be minimized to an extreme level, the weight of $\mathcal{L}_{IED}$ should be set to a smaller value. We conducted hyper-parameter analysis for W_{IED} on multiple datasets within the range of 0.01 to 1, as shown in Figure 5. The results indicate that the optimal W_{IED} values for NightReID and Night600 are 0.1 and 0.5, respectively, which align with the distinct brightness characteristics of each dataset.

3 Visualization

Class Activation Map

To further validate the effectiveness of our proposed method, we employed Grad-CAM (Selvaraju et al. 2017) for visualization, as shown in Figure 6. The Input represents the original image, Enhance shows the image enhanced by the IED module, and BASE, IED, and DDA denote the activation maps for the baseline and branches utilizing IED

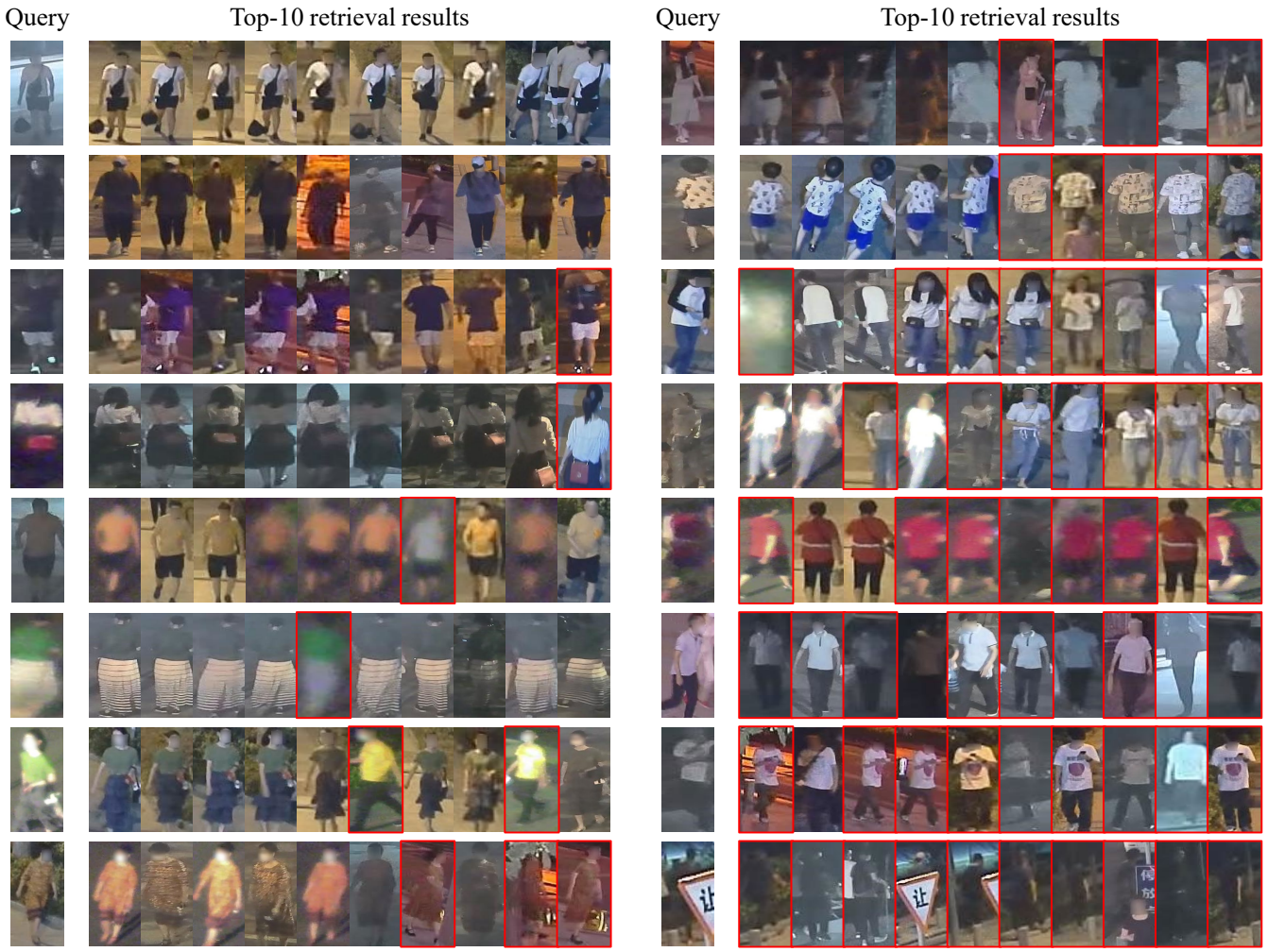


Figure 7: Sample retrieval results from the NightReID dataset.

and DDA modules, respectively. The visualizations demonstrate that our proposed method significantly enhances feature extraction for nighttime Re-ID images. Moreover, the two branches enhanced by separate modules exhibit focused attention on different regions of interest, thereby further improving the overall discriminative power of features.

Retrieval Results

To better illustrate the challenges posed by the NightReID dataset, we present retrieval results for some of the challenging samples in the dataset, as shown in Figure 7. Each query is followed by the top 10 retrieval results, with incorrect matches outlined in red. These retrievals underscore the significant challenges posed by the NightReID dataset, including complex lighting conditions, diverse environments, and varied weather conditions. These challenges often cause different individuals to look similar, while the same individual may appear different across images. Such issues are more prevalent in the NightReID dataset compared to daytime Re-ID datasets. The false retrieval results depicted in the figure further validate these challenges associated with the NightReID dataset.

4 Limitations

The proposed DDA module adjusts the overall data distribution, which may not be well-suited for models requiring additional semantic or posture information extracted from the original data.

5 Privacy Considerations

We clarify that the NightReID dataset adheres to strict privacy standards and is intended solely for academic research. All data were collected under institutional authorization, with images sourced from public surveillance cameras. To safeguard individual privacy, all facial information in the images has been blurred, ensuring the exclusion of any personally identifiable information (PII). Moreover, access to the dataset is restricted to academic research. Users are required to sign an academic agreement, and any commercial or unethical use of the dataset is strictly prohibited.

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